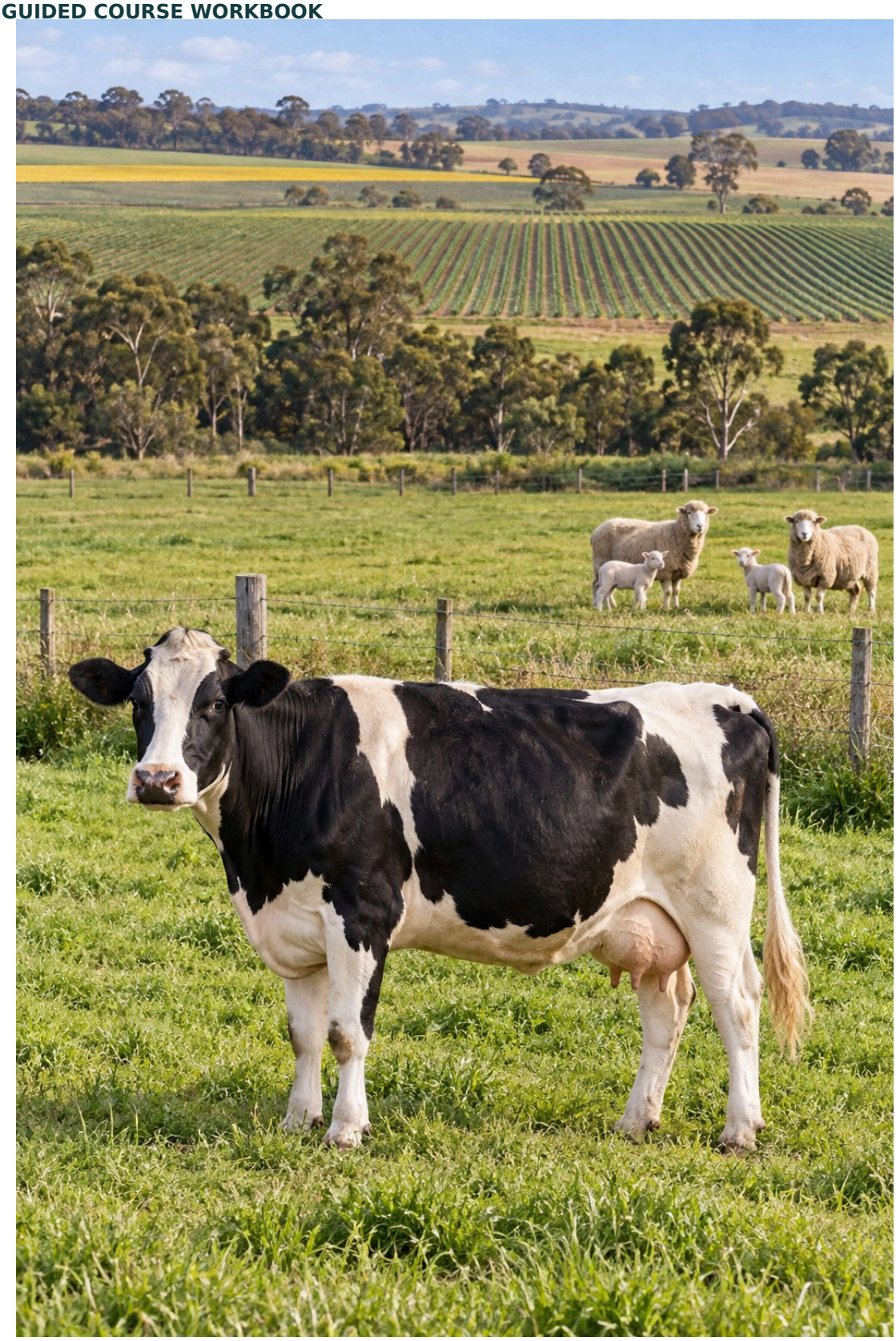
YEAR 10 / GUIDED COURSE COMPANION

Year 10 Agriculture

THE GUIDED COURSE WORKBOOK



Observe, explain, compare and evaluate through the guided course's 20-module sequence.

Name _____ Class _____ Teacher _____ Year ____

Systems · Cropping · Prime lamb · Dairy
Illustrative cropping, prime lamb and dairy context; not the school farm.

50 A4 pages · 25 double-sided sheets · Guided course companion

START HERE / CONTENTS AND LEARNING ROUTINE

Your course companion

Follow the current website sequence. Each module pairs three condensed readings with selected written evidence and a quick check. Application pages bring related ideas together through calculations, investigations, drawing and practical records.

Course strand	Modules and student pages
Systems and management	1–5: pages 3–12 • evidence and checks 13–14
Pastures and cropping	6–10: pages 15–24 • finance and folio 25–26
Prime lamb production	11–15: pages 27–36 • pedigree and folio 37–38
Dairy farming	16–20: pages 39–48 • anatomy and review 49–50

Build a useful response

Explain links a cause or relationship to an outcome. Compare uses the same criteria for each option. Evaluate weighs evidence, limitations and trade-offs before reaching a judgement. Start in the ruled area; continue developed responses, calculations, research and diagrams in your class folio, labelled with the source section ID.

What is included

All 60 named readings are taught in condensed form. The book selects 60 of the website's 102 written tasks and prints 24 representative source multiple-choice questions. The full 600-question bank, all remaining tasks, detailed readings, videos and Busy Work activities remain online. These selections keep meaningful handwriting and drawing space within 50 pages.

Use current instructions

The website's current 2026 AG5 sequence is followed. Current teacher notices control formal assessment requirements, dates, marks and submission. Local access, animal handling, breeding, chemical use, sampling, monitoring and care procedures remain teacher-controlled. Current prices and industry statistics require dated verification.

Course website:
stevencowell.github.io/Year-10-Agriculture-Guided-Course/

Source revision c50916f, retrieved 2 October 2026 • Print A4, double-sided on the long edge: 50 pages on 25 sheets.

MODULE 01 / SYSTEMS AND MANAGEMENT

Regions and diverse enterprises

Regions and enterprise suitability

An enterprise produces agricultural products or services. Suitability depends on interacting conditions: rainfall timing and reliability affect growth, temperature influences performance, and soils control drainage, nutrients and roots. Trafficability is how readily land supports movement and machinery. Plains may support cropping, while pasture on land less suited to cultivation may support grazing.

Market access connects products with buyers, processing and transport. Infrastructure, prices, climate variability, pests, disease and water pressures can challenge production despite suitable land. Crop rotation supports soil health; rotational grazing manages pasture use across areas. Vineyards need long-term establishment. Explain why conditions support an enterprise and what challenges remain, rather than ranking regions automatically.

Goats, bees and pigs

Goats, bees and pigs broaden agriculture. Goats supply milk for dairy products, leather and meat: chevon comes from adults, capretto from young goats. Cashmere is soft undercoat fibre; mohair comes from Angora goats; cashgora combines their characteristics. These outputs serve different consumer needs.

Bees provide pollination, transferring pollen between flowers to support seeds or fruit. Queens lay eggs, workers gather food and rear young, and drones mate with queens. Bee decline could affect food production, availability, economic activity and health. Pigs supply fresh pork and processed bacon, ham, sausages and smoked pork. Direct sales, contracts or consignment, online auctions and saleyards connect producers and buyers; current conditions need verification.

Diversification decisions

Diversification combines enterprises, products or services to spread reliance on sources of value. Goat milk, meat, fibre and leather, bee products and pollination, and fresh or processed pork show different product streams. More streams can bring extra costs, complexity and management, so additional enterprises are not automatically better.

Regional suitability tests climate, soils, water and resources; market access tests buyers and infrastructure. Sustainability involves trade-offs between soil, water, production and management. Rotation, grazing strategies and technology must fit the location. A conditional recommendation compares benefits and limitations and identifies uncertainties. Current demand, costs and conditions need verification because diversification guarantees neither profit nor sustainability. Evidence and long-term planning remain essential.

Suitability is a connected judgement

Physical conditions	Economic practicality	Remaining challenges
Climate, soils, water and landform	Buyers, processing and transport	Variability, pests, disease and resources

Concept organiser based on this module's readings.

MODULE 01 / SYSTEMS AND MANAGEMENT

Explain and check

Written evidence

Begin your explanation here. Continue the complete source response in your class folio labelled M01-S01. Include every comparison or reasoning step requested.

Explain why some Australian agricultural regions face more production challenges than others.

Written evidence

Begin your explanation here. Continue the complete source response in your class folio labelled M01-S02. Include every comparison or reasoning step requested.

Explain how goat, bee and pig enterprises each make a distinct contribution to agriculture.

Written evidence

In five lines, explain spread reliance, identify two added demands or risks, and connect evidence and long-term planning to a justified decision.

Explain why diversification should not be described as a guaranteed way to improve profit or sustainability.

Quick check

How can crop rotation support long-term cropping systems?

- ○ A. It makes every crop grow in every climate.
- ○ B. It removes the need to monitor pests and soils.
- ○ C. It guarantees that products will be sold overseas.
- ○ D. It can support soil health and reduce nutrient deficiencies or disease build-up.

MODULE 02 / SYSTEMS AND MANAGEMENT

Organisms and integrated pest management

Beneficial and harmful organisms

An organism's effect depends on its production relationship. Rhizobium forms a symbiosis with legume roots: it fixes atmospheric nitrogen into a usable form and receives carbohydrates. This mutually beneficial exchange supports fertility and may help later crops. Ladybirds provide biological control by preying on aphids, rather than directly feeding or fertilising crops.

Salmonella is a harmful bacterium associated with poultry and food safety; contamination can affect production and the food chain. Cattle ticks are external parasites, feeding on livestock blood while harming their hosts. They can reduce health and productivity and may transmit disease. Classify organisms by identifying their interactions and production outcomes, rather than assuming that every bacterium or invertebrate is harmful.

Six integrated pest management strategies

Integrated pest management, or IPM, combines controls for economic viability, environmental responsibility and social acceptability. Cultural control changes practices or production conditions. Physical or mechanical control uses barriers, traps or removal. Biological control uses predators, parasites or pathogens. These categories act differently even when addressing the same pest.

Chemical control uses substances as one option. Genetic control uses resistant varieties or breeds; regulatory control uses rules such as quarantine or movement restrictions. Combining suitable categories reduces reliance on one response: changed conditions, barriers and beneficial organisms can support one another. The pest and production context determine suitability. Exact products, rates, biosecurity requirements and local practical procedures need teacher confirmation.

Combining controls and planning for failure

An IPM response begins with evidence identifying the organism or production issue. A justified response combines at least three categories with clear roles. Barriers provide physical control, changed practices provide cultural control, and ladybirds feeding on aphids provide biological control. Environmental effects and beneficial organisms also influence the appropriate mix.

Barriers may leave existing pests unaffected; predators may act slowly or depend on conditions. A contingency plans for failure. Review evidence about identification, suitability and changed conditions, then strengthen another category or replace the failed control. Explain why the revised combination fits the problem instead of promising success. Products, rates, treatment sequences and local procedures remain Teacher to confirm.

Begin with evidence, then compare responses

Identify	Monitor	Compare categories	Review
Organism and role	Pattern and change	Benefits and limitations	Outcome and new evidence

Concept organiser based on this module's readings.

MODULE 02 / SYSTEMS AND MANAGEMENT

Explain and check

Written evidence

In five lines, name one beneficial and one harmful organism, identify their partners or hosts, and link each relationship to its production effect.

Compare one beneficial organism and one harmful organism from this section. Explain each organism's relationship within the production system and its effect on production.

Written evidence

Begin your explanation here. Continue the complete source response in your class folio labelled M02-S02. Include every comparison or reasoning step requested.

Explain the six integrated pest management strategy categories and show how each acts differently.

Written evidence

Begin your explanation here. Continue the complete source response in your class folio labelled M02-S03. Include every comparison or reasoning step requested.

A harmful organism is affecting a production system. Propose and justify an IPM response using at least three different control categories, then explain how the plan could change if one strategy failed.

Quick check

Why is Salmonella considered harmful in agricultural production?

- ○ A. It improves soil fertility
- ○ B. It supports nitrogen cycling in legumes
- ○ C. It preys on aphids
- ○ D. It can affect poultry production and food safety

MODULE 03 / SYSTEMS AND MANAGEMENT

Livestock data and marketing

Reading livestock graphs

Read a livestock graph's title, axes, units and full period. The horizontal axis commonly shows time; the vertical axis shows quantity. Units distinguish animals, thousands and percentages. A trend is the overall direction; fluctuation means rising and falling; a peak is the highest value. Short-term changes may occur within an overall increase or decrease.

Observation states what the graph shows; explanation proposes why. Seasonal conditions, feed or demand may influence numbers, but graphs cannot establish cause. Use cautious wording and supporting evidence. The supplied September 2021–September 2023 data and September 2023 ABS release are historical; their figures and conclusions require checking before being called current.

Comparing selling strategies

Direct selling connects producers with customers, providing presentation control and feedback but requiring time, customer access and communication. Processor contracts arrange supply in advance, supporting planning stability through agreed requirements. Producers must meet standards and conditions, trading some flexibility for predictability.

Online selling extends buyer reach and purchasing convenience but needs digital capability, accurate descriptions and trust. Delivery logistics still determine whether products reach buyers appropriately. A trade-off links a benefit with a limitation. Compare channels through control, stability, reach, workload, standards and delivery, using the producer's goals and resources. Wider reach alone cannot establish suitability, and no channel guarantees profit. Explain whether the producer can meet the chosen channel's demands.

From trends to marketing decisions

Evidence is the change a graph directly records, with units and dates. An inference cautiously connects that observation to an enterprise. Uncertain supply may make flexibility important, whereas predictable planning may favour stability. The graph cannot establish cause, predict prices or choose a selling channel by itself.

Uncertainty includes unknown causes, changing markets and whether an older trend will continue. Compare buyer reach, workload, requirements, transport and trust before making a conditional recommendation. Explain benefits and limitations, then identify current prices, contract conditions, buyer needs and likely supply requiring verification. A justified decision connects observation, inference and recommendation while keeping evidence limits visible. Additional information strengthens the judgement without creating certainty.

A graph describes change; it does not prove cause

Measured observation	Possible explanation
Read title, axes, units and full period	Check supporting seasonal, feed or market evidence

Concept organiser based on this module's readings.

MODULE 03 / SYSTEMS AND MANAGEMENT

Explain and check

Written evidence

Use the hypothetical livestock graph you create on page 13. Describe its units, period, direction and fluctuations without claiming a cause.

Describe the trend in a supplied livestock graph without explaining why it occurred.

Written evidence

In five lines, judge online reach alongside information quality, digital capability, trust and delivery demands; make your conclusion conditional on producer resources.

Explain whether online selling would suit a producer who wants to reach buyers beyond the local area.

Written evidence

Use five lines to distinguish one graph-based fact, one contextual inference and one uncertainty, then identify information that would strengthen the recommendation.

Explain the difference between evidence, inference and uncertainty in a marketing recommendation.

Quick check

What is a likely trade-off of selling through a processor contract?

- ○ A. No need to meet product requirements
- ○ B. Direct feedback from every final customer
- ○ C. Reduced flexibility because agreed conditions must be met
- ○ D. Unlimited freedom to change buyers at any time

MODULE 04 / SYSTEMS AND MANAGEMENT

Surveys and value adding

Clear and unbiased market surveys

A market survey collects preferences, habits and needs for a clear purpose. Its target group is the customer type it aims to represent. Neutral wording avoids steering responses; praising the product introduces bias. Distinct options, including “other” where useful, let respondents report views accurately.

A prediction is an expectation before responses, not a result. Comparing it with collected evidence may expose mistaken assumptions. A sample is the surveyed group: small samples may miss views, while unrepresentative samples may overemphasise one customer type. Conclusions describe those respondents rather than all consumers. Write clear questions relevant to the intended group. Respondent numbers, distribution and local collection requirements remain Teacher to confirm.

Value adding agricultural products

Value adding transforms a raw product through processing, packaging or presentation to meet consumer needs. Potatoes may become chips, bake kits or flour; milk may become cheese, butter or ice cream; flowers may become bouquets. Product differentiation creates a meaningful difference in form, use or presentation that customers must value.

Convenience, broader uses and sometimes longer shelf life, the time available for suitable use or sale, may create selling opportunities. Suitable produce not sold raw may gain another use, reducing waste where requirements are met. Compare benefits with processing, storage, transport and selling costs. Higher price or profit is not guaranteed, and practical food-processing and safety details require confirmation.

Consumer evidence, promotion and value

Consumer evidence comes from responses, not predictions. Patterns may suggest a target audience, the group likely to value particular benefits. Selling points are features selected for promotion with supporting evidence. Interest in convenience, taste, ethics or sustainability does not prove the product’s health, environmental or welfare claims.

An evidence-to-decision chain states the response, interprets a need, selects a matching selling point or value-added form, then identifies uncertainty. Convenience interest may justify investigating an easier-to-use form. Production, storage, transport and selling must remain practical. Small or unrepresentative samples limit conclusions to respondents. Connect audience, needs, message and presentation, but seek separate evidence for product claims and never guarantee demand or profit.

Classroom evidence case

Hypothetical class survey: respondents prefer locally produced cheese and clearer package information. This small convenience sample came from one class. No sales, profit or whole-market data was collected. Use the stated preferences as evidence, separate a possible promotional inference, and identify claims that would require other information.

MODULE 04 / SYSTEMS AND MANAGEMENT

Explain and check

Written evidence

In five lines, state the question’s purpose, write one neutral question for its target group, and provide distinct options without praising the product.

Write one neutral survey question about factors that influence the purchase of an agricultural product.

Written evidence

In five lines, name one raw product and changed form, explain the enhancement generally, and link two possible benefits without promising profit.

Explain how one raw agricultural product could be value added and identify two possible benefits.

Written evidence

In five lines, distinguish surveyed preferences from a cautious inference, identify one claim needing separate evidence, and explain the sample’s limits.

Explain why a promotional recommendation must separate evidence from unsupported claims.

Quick check

What does product differentiation mean?

- ○ A. Reducing the number of possible uses for a product
- ○ B. Making a product meaningfully different from competing products
- ○ C. Guaranteeing that one product controls the market
- ○ D. Selling every agricultural product in the same form

MODULE 05 / SYSTEMS AND MANAGEMENT

Alternative systems and test preparation

Comparing plant production systems

Glasshouses control temperature, humidity and light but need setup, energy and management. Organic production avoids synthetic chemicals and uses natural inputs, with soil-health and biodiversity aims.
Biodynamic production shares features but adds composting practices and lunar-cycle ideas. Descriptions do not prove environmental performance or certification.

Worm farming processes suitable organic waste into castings and liquid fertiliser, providing soil inputs or saleable outputs. It may reduce waste but needs resources, management and reliable use or markets. Compare inputs, control, costs, outputs and limitations. Glasshouse control brings resource demands; organic and biodynamic systems may face yield and certification challenges. Compare enterprise aims and trade-offs; local operation needs teacher confirmation.

Animal production systems and trade-offs

Rangeland beef uses pasture across larger areas and depends on land and seasons. Feedlots confine cattle with controlled high-energy feeding, improving input control but requiring feed, infrastructure and close management. Both need environmental, welfare and cost evaluation rather than judgement from land area alone.

Caged layers use enclosures efficiently, with controlled feeding and egg collection; confinement creates welfare concerns and concentrated waste needs management. Free-range layers have outdoor access but need more land and outdoor management. Consumer interest may influence choice without guaranteeing sales or superiority. A production trade-off balances benefits against resource, welfare or environmental limitations. Verify current standards, requirements and costs before practical decisions.

Topic-test preparation: command terms

Topic-test preparation connects definitions, examples, causes, effects and trade-offs. Describe means identify a concept and features. Explain means show why or how through cause and effect. Combining IPM methods may reduce dependence on one control because their actions differ. Relevant examples make explanations precise.

Compare means identify similarities and differences using shared criteria. Rangeland and feedlot systems both produce cattle but differ in land, feeding and control. Identify the concept, give evidence, add necessary causal links, and answer the question. An explanation can use “This is because” for a reason and “This leads to” for an effect. Check every sentence’s relevance. Test dates, format, marks and local requirements remain Teacher to confirm.



Figure 2. Source cattle contexts: space and infrastructure are visible; welfare and efficiency need further evidence.

MODULE 05 / SYSTEMS AND MANAGEMENT

Explain and check

Written evidence

In five lines, link suitable waste to worm-farm outputs, explain one benefit and one limitation, and mark local operation and safety Teacher to confirm.

Explain how commercial worm farming can contribute to a plant production enterprise.

Written evidence

Begin your explanation here. Continue the complete source response in your class folio labelled M05-S02. Include every comparison or reasoning step requested.

Compare caged and free-range layer systems and explain how consumer demand may affect the decision.

Written evidence

In five lines, explain one value-adding change with a reason, possible effect and example; use causal links and avoid guaranteed-profit claims.

Write an explain response showing why a producer might value-add an agricultural product.

Quick check

What is essential in a compare response?

- ○ A. Similarities and differences organised through shared criteria
- ○ B. A claim that one option is always superior
- ○ C. A separate definition of only one option
- ○ D. A list of facts with no connection between them

SYSTEMS / DATA AND MARKET INVESTIGATION

Read evidence and design a survey

A graph records what changed. A survey gathers views from a defined group. Neither automatically proves why consumers or producers behave as they do.

Worked observation

Hypothetical classroom livestock record, in animals: Week 1 = 12, Week 2 = 15, Week 3 = 13, Week 4 = 18. The count fluctuates but finishes six animals higher. The record alone does not show a cause.

Graph the four hypothetical counts. Include a title, labelled axes, animals as the unit and an even scale.

State one measured pattern and one possible explanation. What evidence would be needed to check that explanation?

Prepare a neutral market question

Purpose and intended respondents	Question and response options

Explain how your wording avoids leading respondents. Identify one limitation of the survey sample.

TERM 1 / KNOWLEDGE AND APPLICATION

Check systems and management

Choose one best answer for each source question. Then transfer your understanding to the short evidence task.

1. Why is a promising product not automatically a strong diversification choice?

- ○ A. The region, infrastructure or likely buyers may not support the enterprise
- ○ B. Diversification decisions should ignore market access
- ○ C. Products have no role in agricultural decisions
- ○ D. Every promising product causes poor soil health

2. Which action is an example of regulatory control?

- ○ A. Installing a trap
- ○ B. Selecting a resistant variety
- ○ C. Introducing a predator
- ○ D. Applying quarantine or movement restrictions

3. Which option best completes an evidence-to-decision chain?

- ○ A. Observed response, cautious interpretation, matched decision and stated uncertainty
- ○ B. Survey title, respondent names, platform choice and fixed price
- ○ C. Prediction, guaranteed conclusion, advertisement and profit claim
- ○ D. Product idea, unsupported claim, broad audience and no limitations

4. What is the main role of a commercial worm farm?

- ○ A. To regulate glasshouse temperature
- ○ B. To certify organic farms
- ○ C. To turn suitable organic waste into castings and liquid fertiliser
- ○ D. To replace all soil-management practices

Make an evidence-based choice

A producer is considering a value-added product. Identify useful customer evidence, one production-system consideration and one sustainability trade-off.

MODULE 06 / PASTURES AND CROPPING

Financial analysis in cropping

Costs, income and gross margin

Fixed costs do not change directly with output: machinery, fencing, sheds and insurance are source examples. Variable costs change with the crop or season, including seed, fertiliser, spray or herbicide, fuel and labour. Classify costs by their behaviour in context, not only their names. Add applicable variable costs for comparison.

Expected income per hectare equals yield in tonnes per hectare multiplied by price in dollars per tonne. Either can change expected income. Gross margin subtracts total variable costs from income, showing the amount left before fixed costs. It compares crops but is not whole-farm profit. Check units, context and included costs before interpreting this expected financial result.

Calculating the wheat gross margin

Gross margin subtracts variable costs from income for the same area. Income per hectare is yield (t/ha) × price ($/t); tonnes cancel, leaving $/ha. The supplied wheat learning example uses 4 t/ha × $300/t = $1,200/ha. These assumptions are not verified current prices or local yields.

The costs $80, $150, $70 and $50 per hectare total $350/ha. Therefore $1,200/ha − $350/ha gives $850/ha gross margin. A reverse check adds $850/ha + $350/ha to recover $1,200/ha income. Recheck multiplication, totals and consistent area units, including only supplied costs. Gross margin is not whole-farm profit because fixed costs and other farm expenses are outside the calculation. Interpret it as one comparison tool.

Comparing crop scenarios

The barley, wheat, canola, lucerne hay and oats scenarios compare yield, price and variable costs. Gross margin is income remaining after variable costs. Rank the supplied margins, treating all figures as learning data, not current market or local-farm facts. Changed assumptions may reverse the ranking.

A sensitivity test changes one assumption: reduce price by 10%, recalculate income and margin, and retain original variable costs. Identify the largest costs to explain differences. Drought-related lower yield would also reduce income and margin; do not invent a yield. A conditional recommendation explains the stronger option under both comparisons and what could change it. Fixed costs and whole-farm profit remain excluded.

Keep financial measures distinct

Income	Gross margin	Whole-farm profit
Yield × price	Income − variable costs	Includes fixed costs and other expenses

Concept organiser based on this module's readings.

MODULE 06 / PASTURES AND CROPPING

Explain and check

Written evidence

In five lines, define how both cost types behave, give two source examples of each, and explain why classification depends on production context.

Explain the difference between fixed costs and variable costs in a cropping enterprise, using source examples.

Written evidence

Use the scenarios and calculation organiser on page 25. Show compact working here; continue the price-change comparison beside that organiser.

Calculate and explain the gross margin for the supplied wheat learning scenario.

Written evidence

Use the scenarios and calculation organiser on page 25. Show compact working here; continue the price-change comparison beside that organiser.

Compare two supplied crop scenarios and recommend one option under the stated learning conditions.

Quick check

A student calculates income in $/ha but uses a total farm cost rather than a cost per hectare. What is the main problem?

- ○ A. The yield must be doubled
- ○ B. The units are inconsistent
- ○ C. Gross margin requires addition
- ○ D. The market price is too low

MODULE 07 / PASTURES AND CROPPING

Crop monitoring and evidence

Crop indicators and patterns

Monitoring records observations and data regularly. An indicator is a clue, not a diagnosis. Plant population describes plants and gaps; distribution shows where signs occur. Record colour, damage, weeds, growth stage and moisture. Yellowing may indicate nitrogen deficiency or waterlogging; purple colouring may suggest phosphorus issues. Similar signs can have different causes.

Chewed edges may suggest feeding damage; spots indicate a possible health issue; dense weeds show competition. Wilting or dryness may indicate moisture pressure. Compare affected and unaffected areas over time: a localised low-area pattern differs from crop-wide signs. Describe observations, uncertainties and further evidence needed. Monitoring supports investigation; local sampling, equipment use and responses remain teacher-controlled.

Monitoring tools and evidence

Field observations from crop walks show soil, leaf colour, gaps and damage closely. Drone or NDVI images reveal aerial patterns such as patchy growth. Satellite data shows broader vigour, rainfall and greenness. These tools identify patterns for investigation, not diagnoses. Their scales and evidence types differ.

Soil-moisture sensors track water within soil layers. Crop models and apps predict development or potential yield using information and assumptions. Choose tools for the evidence needed: broad imagery may locate an area, while closer observations add detail. Compare images, readings, models and field observations over time. State the pattern, its source and missing information before a conclusion. Local tool use remains teacher-controlled.

Prioritising monitoring investigations

Regular monitoring reveals developing problems and may prevent unnecessary inputs. A priority is the issue most justified for earlier investigation. Compare extent, how much crop appears affected, change over time, possible growth or yield effects, and evidence quality. Dramatic appearance alone does not establish priority.

Yellowing, chewed leaves, weeds, wilting and spots remain indicators, not confirmed causes. A widespread expanding pattern may deserve attention before a small stable patch. Explain the observations supporting that judgement, what could happen if it continues and what evidence is missing. Targeted investigation may support resource efficiency and soil and plant health while reducing unnecessary input losses. Local thresholds, actions and practical requirements remain teacher-controlled.

Classroom evidence case

Hypothetical crop records compare first and later observations. North: yellowing and gaps spread from a small patch; west remains green. Centre: dense unwanted plants expand across a quarter of the crop. East: chewed edges affect a few plants without change. Drone images show area patterns; crop-walk notes describe leaves. Causes and moisture conditions remain unconfirmed.

MODULE 07 / PASTURES AND CROPPING

Explain and check

Written evidence

In five lines, compare visible indicators across areas or times, describe the pattern, identify missing evidence, and state that diagnosis remains unconfirmed.

Analyse a supplied crop-monitoring case and explain the pattern shown by the observations.

Written evidence

In five lines, compare two tools' evidence scales, show how their observations complement each other, and identify time comparisons and missing evidence.

Compare two crop-monitoring tools and explain how their evidence could be used together to investigate a crop-health pattern.

Written evidence

Begin your explanation here. Continue the complete source response in your class folio labelled M07-S03. Include every comparison or reasoning step requested.

Compare several supplied crop-monitoring cases, select one priority for further investigation and justify your choice.

Quick check

What do soil-moisture sensors track?

- ○ A. Pest species from appearance alone
- ○ B. Water within soil layers
- ○ C. Current market prices
- ○ D. Leaf colour from the air

MODULE 08 / PASTURES AND CROPPING

Weeds, pests and disease evidence

Weeds and competition

A weed grows where it is unwanted. Competition occurs when weeds and useful plants seek light, nutrients and water, possibly reducing yield or pasture quality. Annual ryegrass, capeweed and thistles are examples, not identification evidence. Establish the plant, crop or pasture, distribution, growth stage and conditions.

Distribution means where weeds occur: scattered plants differ from dense widespread growth. Broad categories include rotation, competitive crops or pastures, herbicides with resistance management and grazing-pressure control. Describe competition, available evidence and gaps before conditionally discussing a category. Appearance cannot justify identity or management. Local actions need teacher confirmation, while formal collection records must come from authentic authorised student evidence, not guessed identities, dates or locations.

Pests and monitoring evidence

Pests damage plants directly. Aphids, red-legged earth mite and armyworm are examples, not identifications of ambiguous specimens. Monitoring gathers decision evidence: damage, possible pest presence, distribution and change. Chewed leaves, discolouration and reduced growth remain indicators. Distribution shows where damage occurs; widespread increasing damage differs from a few stable affected plants.

A threshold is a decision category, not an invented numerical trigger. Broad approaches include biological control using living organisms, targeted insecticides and avoiding unnecessary spraying. Separate observation from suspected explanations and identify further evidence, such as pest presence or unaffected-plant comparisons. Discussion stays conditional because appearance cannot establish cause or treatment.
Local actions and genuine formal collection evidence remain teacher-controlled.

Diseases and integrated management

Disease may involve fungi, bacteria or viruses, and may spread through wind, moisture or infected residue. Rust, root rot and leaf spots are examples. Observation records visible signs; diagnosis identifies a disease or cause and needs further evidence. Describe distribution and change without naming an unconfirmed pathogen because similar signs can have different causes.

Integrated management considers suitable broad categories together using evidence and context: resistant varieties, rotation, fungicides and reduced residue carryover. Resistance and rotation may warrant investigation, but every category does not fit every case. Identify possible pathways, missing evidence and limits before considering responses. Products, rates, timing and procedures require teacher confirmation. Formal collection evidence must be authentic.

Classroom evidence case

Hypothetical crop observations: dense unwanted plants occupy the centre among crop plants. East has chewed leaf edges and reduced growth, affecting more plants later; west remains unaffected. Leaf spots occur in two northern areas, then appear in additional areas. Plant identities, pest presence, growth stages, weather and residue conditions are unconfirmed. No treatment has been selected.

MODULE 08 / PASTURES AND CROPPING

Explain and check

Written evidence

In five lines, describe distribution and competition, link possible crop effects, identify missing evidence, and conditionally discuss one teacher-confirmed broad management category.

Using a supplied hypothetical case, explain how weed competition could affect a crop or pasture and identify one broad management category for further consideration.

Written evidence

In five lines, describe damage, distribution and change, identify a limitation and further evidence, and discuss one broad category without prescribing treatment.

Interpret a supplied hypothetical crop-monitoring case that includes visible leaf damage and changes over time.

Written evidence

Begin your explanation here. Continue the complete source response in your class folio labelled M08-S03. Include every comparison or reasoning step requested.

Interpret a supplied hypothetical plant-health case showing visible leaf spots across several crop areas and explain how integrated management could be considered without diagnosing the cause.

Quick check

What may cause plant diseases?

- ○ A. Machinery and fencing
- ○ B. Only weeds
- ○ C. Fungi, bacteria or viruses
- ○ D. Only low market prices

MODULE 09 / PASTURES AND CROPPING

Sustainability and environmental management

Sustainability and environmental risks

Sustainable agriculture meets present needs without compromising future generations, balancing environmental health, economic viability and social responsibility. Trade-offs connect these dimensions. Bare ground or overgrazing may increase erosion, soil loss removing fertile topsoil. Excess fertiliser may contribute to runoff, pollution and algal blooms; continuous cropping or fertiliser use may contribute to soil acidification and reduced growth.

Rising water tables may contribute to salinity and poor establishment. Monocultures rely on one crop or pasture type, potentially reducing biodiversity and increasing biological pressure. These relationships are risks, not paddock diagnoses. Identify the practice, possible environmental effect and production consequence; consider economic and social responsibilities, evidence limits and further information before concluding damage occurred.

Grazing, ground cover and recovery

Ground cover is plant material protecting soil. Declining cover or overgrazing may increase erosion, topsoil loss and fertility risks, but one photograph cannot establish cause or damage. Compare observations over time: increasing bare ground suggests pressure; renewed cover may suggest recovery, improving pasture cover after earlier pressure.

Rotational grazing manages pressure across areas to support recovery and cover. It is a principle, not a universal sequence, duration or stocking rate. Moving sheep every 5–7 days is illustrative source learning data, never a local rule. Describe the pattern, possible soil or production risk and missing evidence. Local decisions need confirmed conditions and teacher-controlled supervision, welfare, WHS and practical requirements.

Rotation, tillage and precision inputs

Rotation changes the crop or pasture sequence, potentially reducing monoculture-related biodiversity, pest and disease pressures. Reduced tillage limits disturbance; stubble retention keeps protective plant material at the surface. Together they may protect soil and reduce erosion risk, without proving damage or guaranteeing every soil problem will be solved.

Precision application directs inputs towards evidence of need, potentially reducing waste and fertiliser runoff, pollution or algal blooms. It does not mean less input everywhere. Identify the hypothetical pressure, compare relevant strategies and connect possible environmental and production benefits. Weigh limits before a conditional recommendation. Local sequences, products, rates, calibration, machinery settings and handling procedures remain Teacher to confirm.

Classroom evidence case

Hypothetical production records show the same crop repeatedly grown. At observation one, plant cover is extensive; at two, bare areas expand; at three, some cover returns. Weed pressure is increasing. Soil loss, fertility change and causes are unconfirmed. The business wants continued production while protecting soil and future users. Local grazing and field-operation details are unknown.

MODULE 09 / PASTURES AND CROPPING

Explain and check

Written evidence

In five lines, link the observed practice to one possible environmental and production risk, include three sustainability dimensions, and identify evidence limits.

Interpret a supplied hypothetical agricultural scenario by identifying one possible sustainability risk and explaining its possible production consequences.

Written evidence

In five lines, compare cover changes across times, explain one possible risk and broad grazing principle, then state uncertainties and further evidence.

Interpret supplied hypothetical observations showing changes in ground cover across several monitoring times.

Written evidence

Begin your explanation here. Continue the complete source response in your class folio labelled M09-S03. Include every comparison or reasoning step requested.

Compare two broad sustainability strategies for a supplied hypothetical production pressure and recommend which should be investigated further.

Quick check

What possible risk is linked with rising water tables?

- ○ A. Increased topsoil depth
- ○ B. Guaranteed higher yield
- ○ C. Salinity and poor establishment
- ○ D. Automatic disease control

MODULE 10 / PASTURES AND CROPPING

Evaluation and agricultural careers

Reviewing an integrated production system

An integrated review connects evidence across production. Examine plant structures, growth, colour and distribution alongside soil, ground cover and monitoring patterns. Compare areas and times to determine whether change is localised, widespread, stable or increasing. Discuss possible biological pressure only where evidence supports it; indicators do not establish cause.

Sustainability evidence concerns soil, water and biodiversity pressures and broad strategies such as rotation, cover, reduced tillage or targeted inputs. Financial evidence adds income, variable costs and gross margin, the difference between them. Margin excludes fixed costs and is not whole-farm profit; figures are learning scenarios. Trace cautious connections, justify a priority for investigation and identify missing evidence before any conclusion or local response.

Evaluating with four connected criteria

Evaluate productivity, sustainability, profitability and risk management together. Productivity concerns output, using yield, biomass, regrowth or uniformity. Sustainability concerns long-term soil, cover, water and biodiversity. Strong output cannot prove environmental strength: declining cover creates trade-offs despite high yield. Visible indicators support evaluation without diagnosing cause.

Profitability compares income, variable costs and gross margin, which excludes fixed costs rather than showing whole-farm profit. Higher yield can accompany higher costs and weaker margin. Risk management examines climate, biological, market and dependence risks; relying on one crop or market increases vulnerability. Compare supplied learning evidence across all four criteria, explain trade-offs and make a conditional recommendation. New evidence may change the judgement.

Careers, skills and pathway exploration

Career examples include farmer or farm manager, machinery operator, agronomist, agricultural scientist, precision agriculture technician, agricultural consultant, and seed or fertiliser representative. Career exploration links interests with demonstrated skills: monitoring shows observation and pattern recognition; data work supports technology interests. Sustainability, risk and gross-margin comparisons show evaluation and financial reasoning; justified recommendations show communication.

Pathways include TAFE, university, apprenticeships or traineeships, and on-the-job learning. A pathway develops skills, not a confirmed enrolment option. A verification gap is information needing confirmation: providers, qualifications, credentials, entry requirements or availability. Explain why a career and pathway merit investigation using learning evidence. Conclusions remain conditional; examples do not establish current vacancies, salaries or required qualifications.

Classroom evidence case

Hypothetical systems: A yields 4 t/ha, has declining cover, fewer plant types and increasing gaps; income $1,200/ha, variable costs $900/ha, one crop/buyer. B yields 3 t/ha, maintains cover and varied plants; income $900/ha, variable costs $500/ha, diversified crops/buyers. Soil causes remain unknown. Student profile: enjoys technology and observation; demonstrates monitoring-data comparison and justified recommendations.

MODULE 10 / PASTURES AND CROPPING

Explain and check

Written evidence

Begin your explanation here. Continue the complete source response in your class folio labelled M10-S01. Include every comparison or reasoning step requested.

Evaluate a supplied hypothetical crop or pasture system case and identify one priority for further investigation.

Written evidence

Begin your explanation here. Continue the complete source response in your class folio labelled M10-S02. Include every comparison or reasoning step requested.

Compare two supplied hypothetical agricultural systems using productivity, sustainability, profitability and risk management, then make a conditional recommendation.

Written evidence

In five lines, link profile interests and two demonstrated skills to one career and pathway, then identify current details requiring verification.

Use the supplied hypothetical student profile to identify one agricultural career category and one broad pathway category that may be worth further investigation.

Quick check

What does gross-margin work demonstrate?

- ○ A. A guaranteed career pathway
- ○ B. Knowledge of current salaries
- ○ C. Financial reasoning using income and variable-cost evidence
- ○ D. Proof of whole-farm profit

PASTURES AND CROPPING / WORKED EXAMPLE

Cropping gross margins

Use the same area and production period for both options. Gross margin excludes fixed costs, so it is not whole-farm profit.

Source learning example: wheat

Income: 4 t/ha × $300/t = $1,200/ha

Variable costs: seed $80 + fertiliser $150 + herbicide $70 + fuel $50 = $350/ha

Gross margin: $1,200 − $350 = $850/ha

Reverse check: $850 + $350 = $1,200/ha

Hypothetical option B	Yield	Price	Variable costs
Second crop for practice	3.5 t/ha	$320/t	$420/ha

Hypothetical B costs per hectare: seed $100, fertiliser $180, herbicide $90 and fuel $50; total $420.

Calculate option B's income and gross margin per hectare. Show the units and each step.

Reduce both crop prices by 10%, keeping yield and variable costs unchanged. Recalculate both gross margins and compare the result.

Recommend an option using both comparisons. Explain how drought-related lower yield could change the result, without inventing a new yield.

Option B is newly authored practice data, not current market information or a local farm recommendation.

PASTURES AND CROPPING / PRACTICAL FOLIO

Plant collection and monitoring evidence

Use a teacher-approved plant specimen, photograph or supplied example. This learning record supports the Plant Collection and Research task; follow the current formal task notice for its required sections and submission.

Plant or sample reference	Date, place and evidence source

Draw the observed plant or specimen. Label visible structures carefully; add a scale only if it was measured or supplied.

Visible indicator	Location or change over time	Further evidence needed

Explain how one visible structure relates to function. Separate an observation from a possible cause of poor performance.

Monitoring signs are clues, not a student diagnosis.

Sampling, tools, contact, chemicals and local responses remain teacher-controlled.

MODULE 11 / PRIME LAMB PRODUCTION

Breeding systems and pedigree

Planning a prime lamb breeding system

Prime lamb breeding aims for traits suited to meat production, including growth, muscling, sound structure and fertility. A maternal line contributes the ewe's ability to conceive and raise lambs; a sire is the male parent, with meat-focused sires contributing growth and muscling. The source uses Merino, Border Leicester, Poll Dorset and White Suffolk to introduce complementary roles.

Crossbreeding mates different breeds within one species to combine useful traits. A breeding stage can produce crossbred ewes, followed by mating with a meat-focused sire. Start with the enterprise goal and select complementary characteristics, rather than one feature. These genetic choices influence future generations but cannot guarantee identical offspring.

Reading a prime-lamb pedigree

A pedigree is a labelled parent-to-offspring diagram, read along its arrows or connecting lines. Task 3 begins with a Merino ewe and a student-selected variable ram. Their offspring is a first-cross ewe, a female sheep combining characteristics from both parent breeds. She is distinct from either original parent and becomes the female in the second mating.

That first-cross ewe mates with a student-selected terminal sire, the final male parent in this structure. Their prime lamb offspring are not retained as breeding females to continue the pedigree. Keep generations, parents and offspring clearly linked. Breed choices need authorised evidence about characteristics, main product and contribution to the exact role; popularity alone is insufficient.

Reproductive technologies: purpose and limits

Artificial insemination, or AI, accesses a selected sire's genetics through semen for chosen ewes. Embryo transfer, or ET, uses embryos combining a selected donor ewe's and sire's genetics. A recipient ewe supports development and raises the lamb without contributing inherited genetics. Thus AI widens selected sire access, while ET can extend valuable maternal and sire lines.

Both technologies support planned genetic improvement without guaranteeing particular offspring traits. Compare the breeding purpose and genetics sought with costs, specialist expertise and access to suitable animals, genetics and services. ET may be more complex and unsuitable for some enterprises. These are biological concepts, not practical procedures; local decisions and reproductive work remain teacher-controlled.

Two stages, two different sire roles

First mating	Second mating
Merino ewe + selected variable ram → first-cross ewe	First-cross ewe + selected terminal sire → prime lambs

Concept organiser based on this module's readings.

MODULE 11 / PRIME LAMB PRODUCTION

Explain and check

Written evidence

In three or four sentences, connect the enterprise goal with selected traits and their possible influence on future generations.

Explain why planned breeding decisions matter in a prime lamb breeding system.

Written evidence

Describe each mating's parents and offspring, then explain how the first stage supplies the female for the second in four sentences.

Explain how the two mating stages in the Task 3 pedigree are connected.

Written evidence

Write a short comparison identifying the genetic contribution in each technology and the recipient's role, without describing practical steps.

Compare the genetic purpose of artificial insemination and embryo transfer.

Quick check

Which factor is identified as a possible limitation of both AI and ET?

- ○ A. Financial cost and specialist expertise
- ○ B. The need to identify sheep by colour
- ○ C. The inability to use selected genetics
- ○ D. The removal of all breeding choices

MODULE 12 / PRIME LAMB PRODUCTION

Feed and pasture evidence

Nutrients, digestion and energy

Sheep need balanced nutrients and water. Energy from carbohydrates and fats supports movement, warmth, digestion, growth and reproduction. Protein supplies material for tissue growth and repair; clover, lucerne and lupins are source examples. Fibre supports the rumen, a digestive compartment where microorganisms break down plant material. These functions connect; one nutrient cannot substitute for all the others.

Leafy pasture is generally more digestible than old, stemmy feed. Very fibrous material may fill the rumen while providing less useful energy or protein. Vitamins and minerals support processes including bones, immunity and reproduction; water supports digestion and temperature regulation. Compare feed balance and digestibility with production demand. Rations, supplements and local feeding choices require teacher confirmation.

Matching feed to changing requirements

Maintenance feeding supports basic functions and helps sheep hold condition. Growing lambs build tissue, pregnant ewes support developing lambs, and lactation means milk production. These stages increase nutritional demand, so maintenance feed may be inadequate. Animal condition and production stage therefore matter when assessing what feed is suitable.

Quality and quantity must be considered together. Leafy pasture is generally more digestible; legumes may supply useful protein, but a few desirable plants do not establish a paddock's adequacy. Too little high-quality pasture can leave demand unmet, while plentiful stemmy feed may provide insufficient useful nutrition. Supplementation adds feed to fill an identified gap; exact supplements, quantities and local decisions require teacher confirmation.

Pasture sampling and feed evidence

Pasture sampling estimates available feed from selected areas. Representative sampling includes normal variation, avoiding only unusually green, bare or heavily grazed patches. A quadrat provides a consistent area. Changing the sampled area, material or measurement can make results reflect method differences rather than pasture variation. Local dimensions and procedures require teacher confirmation.

Dry matter excludes water, helping compare actual plant material when fresh samples differ in moisture. However, each sample represents only part of a variable paddock. Quantity does not fully reveal quality or likely consumption: abundant stemmy feed may be less nutritious. Combine estimates with pasture condition, animal requirements and condition. Sampling informs decisions; it cannot independently determine local feeding or stocking responses.



Figure 3. Observe representative leaves, stems and roots; these forms neither identify species nor establish feed value.

MODULE 12 / PRIME LAMB PRODUCTION

Explain and check

Written evidence

Use three or four linked sentences to distinguish nutrient functions and explain the limitation of very stemmy feed for sheep.

Explain how energy, protein and fibre perform different roles in sheep nutrition.

Written evidence

Compare the two pasture situations in three or four sentences, explaining how both amount and useful nutrition affect whether demand is met.

Explain how pasture quality and quantity should be considered together when matching feed to sheep requirements.

Written evidence

Explain one sampling bias and one method-consistency issue, then link both to the reliability of a pasture estimate in four sentences.

Explain why representative samples and a consistent method are important when estimating pasture availability.

Quick check

What is the main purpose of maintenance feeding?

- ○ A. To produce the fastest possible growth
- ○ B. To prepare every sheep for lactation
- ○ C. To support basic body functions and maintain body condition
- ○ D. To replace all pasture with supplements

MODULE 13 / PRIME LAMB PRODUCTION

Planning and monitoring

A management calendar as a planning tool

A management calendar links events across a production cycle, the recurring enterprise sequence. Joining influences lambing; lambing affects weaning; feeding, shearing and marketing also connect with other stages. Timing changes affect animal demand, labour, growth and marketing, so preparation should consider these relationships rather than isolated jobs.

Entries can record an event's purpose, resources, observations and follow-up. Pasture availability, feed condition, animal condition and actual timing help compare the plan with results. Review uses that evidence to improve later planning. Calendars remain adaptable because seasons, goals and resources change. Source timings demonstrate organisation, not a universal farm schedule. Exact dates and local procedures require teacher confirmation for the particular enterprise.

Linking decisions across the production cycle

Production stages connect because feed demand, animals' nutritional requirements, changes with reproduction, growth and ewe recovery. Limited pasture before a demanding stage may affect condition and later performance. Joining influences lambing; lambing outcomes affect later growth groups. Weaning separates lamb and ewe nutritional needs, allowing separate consideration of their growth and condition.

Monitoring collects observations over time. Pasture, ewe condition and lamb growth records can reveal whether earlier decisions were followed by expected results. One observation identifies a possible concern; linked records show patterns without proving one cause. Misaligned timing, feed and requirements may reduce growth or recovery and complicate marketing. Review earlier outcomes, present evidence and next-stage demands; local actions require teacher confirmation.

Technology for monitoring and decisions

Electronic identification, or EID, links each animal to its digital record. Weights, growth, lambing history and observations can then be matched correctly. Repeated measurements reveal changes over time that one measurement cannot show. Weighing and scanning records support comparisons across animals, groups and production stages, reducing reliance on memory.

Data quality means accuracy, completeness and consistency. Missing records or incorrect entries weaken conclusions; a pattern does not establish its cause. Technology also has cost, maintenance, training and access limits. Compare digital records with direct observations, seasonal conditions and animal requirements. More data helps only when it informs the actual decision. Technology organises evidence and strengthens human judgement; it cannot replace interpretation or guarantee better outcomes.



Figure 4. Compare visible pasture cover; appearance alone does not rank feed quality, quantity or lamb growth.

MODULE 13 / PRIME LAMB PRODUCTION

Explain and check

Written evidence

Identify two useful calendar records and explain how comparing them with the plan could improve a later review in four sentences.

Explain why records and observations should be included in a management calendar.

Written evidence

Use one hypothetical stage-to-stage link; state the earlier evidence and explain one possible later consequence without recommending local management actions.

Explain how a feed-related decision at one stage of the production cycle could affect a later stage.

Written evidence

Explain one useful contribution of technology and two limits of the evidence, then state why interpretation still needs human judgement.

Explain why technology should support rather than replace human judgement.

Quick check

How should a source example of an annual sheep calendar be used?

- ○ A. As an example of how production stages can be organised and linked
- ○ B. As a replacement for enterprise records
- ○ C. As a fixed schedule that applies to every farm
- ○ D. As proof that local conditions do not matter

MODULE 14 / PRIME LAMB PRODUCTION

Genetics, ethics and welfare

Genetic change: selection and engineering

Genes are DNA sections influencing traits such as growth and structure, but environment and management also affect expression. Selective breeding chooses parents with desirable existing traits and assesses offspring across generations. It works with genetic variation rather than rewriting DNA, and cannot guarantee every offspring's characteristics.

Genetic engineering directly changes DNA through biotechnology, inserting, deleting or altering genes to seek traits. Compare the intended change with reliable measurements, fair comparisons and other possible influences. One observation or promised result cannot prove benefit or harm. Evidence may support a result under particular conditions while leaving uncertainty. A comparison explains how each approach changes traits and evaluates the quality of evidence, without assuming either guarantees success.

Ethical reasoning about biotechnology

Ethical reasoning examines a biotechnology proposal and its claims. Stakeholders are those affected, including producers, scientists, consumers, animals and the environment; group members can disagree. A factual claim can be checked with evidence. A value judgement expresses what someone considers right, fair or important. Uncertainty identifies what remains unknown. Distinguishing these statements improves reasoning.

Compare benefits, harms and unknowns for several stakeholders. Examine measurements, fair comparisons, source reliability and other explanations. A benefit does not automatically outweigh harm; a concern does not prove harm will occur. A justified conclusion can support, oppose or conditionally accept the proposal. Conditional acceptance states the further evidence or safeguards needed. No particular moral position is required.

Welfare observations and husbandry records

Animal welfare concerns condition and experience, not production alone. Husbandry is routine management and care, influencing food, water, shelter and health care. An objective observation records visible behaviour, movement, feeding or condition without guessing causes. Continued growth does not establish that all welfare needs are met.

Record events, context and changes from usual patterns. A sheep remaining apart is observable; claiming disease is an interpretation needing further evidence. Repeated records may reveal persistent changes more clearly than one sign. These records assist a teacher or qualified person in deciding whether assessment is needed. Students describe evidence without diagnosing or prescribing treatment. Local welfare rules, handling, supervision and reporting pathways require teacher confirmation.

An ethical judgement needs evidence

Purpose	Consequences	Alternatives
Why a technology or practice is considered	Production, welfare and wider impacts	Benefits, limitations and responsibility

Concept organiser based on this module's readings.

MODULE 14 / PRIME LAMB PRODUCTION

Explain and check

Written evidence

In three or four sentences, compare how the approaches change traits and explain the main difference in their use of genetic variation.

Compare selective breeding and genetic engineering as ways of changing traits in a population.

Written evidence

Give one brief hypothetical statement of each type and one sentence explaining why distinguishing them improves an ethical argument.

Explain the difference between a factual claim, a value judgement and an uncertainty in an ethical debate.

Written evidence

Write a brief hypothetical observation, then explain why it records visible evidence without establishing a diagnosis or prescribing an animal-care response.

Write an objective welfare observation and explain why it is stronger than a diagnosis.

Quick check

Why should stakeholder groups not be treated as having one shared opinion?

- ○ A. Only scientists are affected by biotechnology
- ○ B. Ethical questions have one correct response for each group
- ○ C. People within a group may have different priorities, values and experiences
- ○ D. Stakeholders cannot understand scientific evidence

MODULE 15 / PRIME LAMB PRODUCTION

WHS and farm business

Hazards, risks and control measures

A hazard can cause harm; risk is the possibility of harm through exposure. A control measure eliminates, reduces or manages the hazard or risk. The hierarchy orders controls as elimination, substitution, engineering, administrative controls and personal protective equipment, or PPE. Elimination removes hazards; substitution replaces them with less hazardous alternatives.

Engineering physically separates people from hazards. Administrative controls rely on instructions, training or supervision. PPE is worn to reduce harm but usually leaves hazards present. Higher controls are preferred because they depend less on behaviour; PPE alone cannot establish safety. Poor controls may affect people, property, environments and animal welfare. Actual ratings, controls, equipment, supervision and procedures require approved local teacher confirmation.

Reasoning through a risk assessment

A risk assessment links task and context, hazard, consequence, risk level, controls and review. A consequence describes possible harm through exposure, distinct from the hazard itself. Records identify those potentially affected. Source categories of low, medium and high require an approved local method, not guesswork.

Controls need their hierarchy level and responsibility recorded. Responsibility identifies who has authority to organise, check or approve the measure; writing a proposal does not establish implementation. Review is needed when the task, surroundings, equipment, animals or people change, or new evidence challenges the assessment. This keeps reasoning connected to current conditions. Students cannot invent ratings or approvals; local matrices, controls, supervision and sign-off require teacher confirmation.

Farm costs, gross margins and SWOT

A budget organises expected income and costs. Variable costs change with production, such as feed supplements; fixed costs generally remain, such as rates. Gross margin is enterprise income minus variable costs. It excludes fixed costs, so cannot establish whole-farm profit. Source income uses number sold multiplied by income per animal; variable costs use number produced multiplied by combined per-animal costs.

Fair comparisons use consistent assumptions, periods and included costs. Increasing income, reducing costs or improving efficiency requires evidence about consequences. SWOT distinguishes internal strengths and weaknesses from external opportunities and threats. Business choices also consider production, welfare, WHS and longer-term effects. Actual classifications, prices and recommendations require teacher confirmation.

Hazard and risk are different

Hazard	Risk assessment
Something with potential to cause harm	Consider consequence, likelihood and controls in context

Concept organiser based on this module's readings.

MODULE 15 / PRIME LAMB PRODUCTION

Explain and check

Written evidence

Use one hypothetical situation to identify the hazard, explain exposure and possible harm, and classify one conceptual control within the hierarchy.

Explain the difference between a hazard, a risk and a control measure using a hypothetical situation.

Written evidence

In three or four sentences, explain what responsibility adds to a written control and identify changes that would justify reviewing the assessment.

Explain why a risk-assessment record must identify responsibility and include review points.

Written evidence

State the gross-margin relationship, identify the included and excluded cost categories, and explain the limit this creates in three or four sentences.

Explain why gross margin and whole-farm profit are not the same measure.

Quick check

Which classification is correct in a SWOT analysis?

- ○ A. All four categories describe external conditions
- ○ B. Strengths and opportunities are internal
- ○ C. Weaknesses and threats are internal
- ○ D. Strengths and weaknesses are internal, while opportunities and threats are external

PRIME LAMB / TECHNICAL EXPLANATION AND DRAWING

Construct a prime-lamb pedigree

A pedigree shows parent-to-offspring relationships. The Task 3 structure has two matings; the first-cross ewe links the generations. Select breeds using authorised evidence and justify each role.

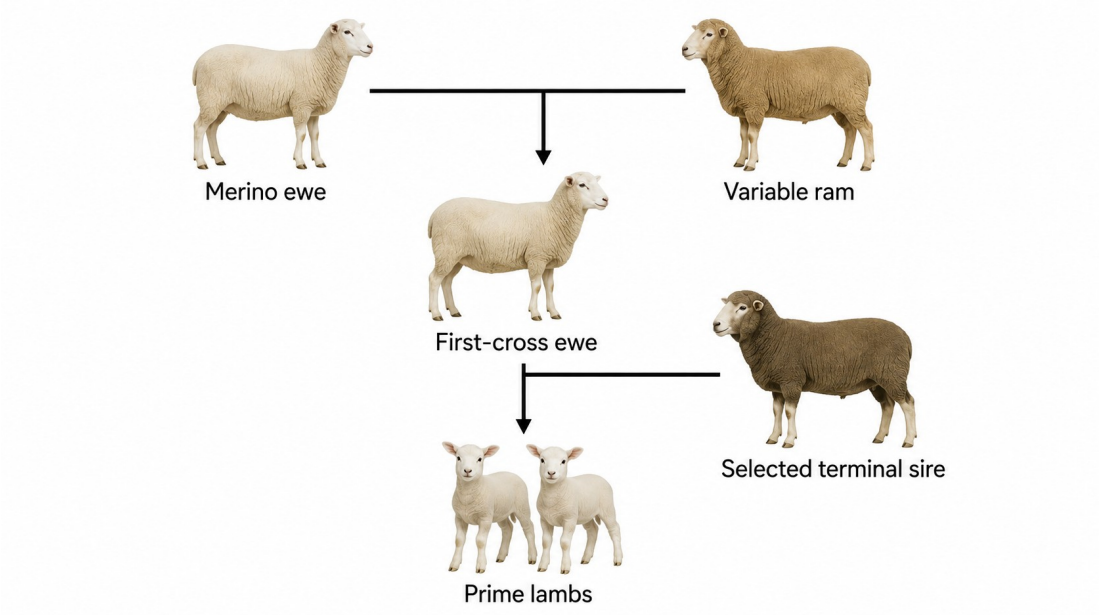


Figure 5. Conceptual breeding relationships, not to scale. The two ram roles are distinct. No variable-ram or terminal-sire breed recommendation is supplied.

Draw your pedigree. Label both parents at each mating, their offspring and each generation. Keep the first-cross ewe distinct from the terminal sire and prime lambs.

Use source evidence to justify your selected variable ram and terminal sire. Explain their different contributions and one limitation.

This page is a learning organiser, not a completed formal submission or permission to conduct breeding procedures.

PRIME LAMB / PRACTICAL FOLIO

Prime-lamb management and risk evidence

Connect production stages with pasture, animal and business evidence. Teacher-provided records and simulated examples are suitable when practical access is unavailable.

Production stage	Feed or pasture evidence needed	Decision or uncertainty

Record a welfare observation and its limits

Date and visible condition	Cautious interpretation	Teacher referral or evidence gap

Analyse a teacher-selected task

Hazard	Possible consequence	Control category to confirm

Evaluate one management choice using welfare, production, costs and risk. Explain a trade-off with reference to evidence.

Do not record handling skills as completed unless actually observed and signed off through the teacher's formal process.

MODULE 16 / DAIRY FARMING

Dairy breeds and external anatomy

Dairy importance and breed selection

Dairy supplies nutrients and supports farming, transport, processing and retail jobs. Selection compares milk yield, the volume produced, with milk composition, including fat and protein. Source tendencies associate Holstein-Friesians with high volume; smaller Jerseys with very high fat and hotter conditions; Guernseys with rich golden milk, high butterfat and protein, and gentle temperament; and Australian-adapted Illawarras with balanced yield and fat.

Ayrshires are good grazers with strong legs; hardy, calm Brown Swiss produce high-protein milk. Temperament means behavioural tendency and matters alongside climate, body size and feed efficiency. Compare breeds with the enterprise goal, feeding conditions and system. No breed is universally best, and tendencies cannot guarantee individual performance. Current industry figures and local choices require confirmation.

External dairy-cow anatomy and function

External anatomy connects structures with function. The muzzle at the head's front gathers feed and water. Eyes provide wide vision; ears support hearing and flick insects. The dewlap assists heat loss. Shoulder and withers indicate condition and muscling; back and loin relate to structural soundness. Deep ribs and barrel support digestive capacity.

The udder produces milk in four quarters with a large blood supply; teats provide milk exits. Hooves support body weight, so poor hoof health may reduce mobility and production. The hock is a hind-leg joint; the tail swats flies. Explain production through function rather than names alone. These relationships do not authorise diagnosis or care; local animal contact remains teacher-controlled.

Using breed and anatomy evidence

A breed tendency is a general characteristic, not an individual guarantee. Holstein-Friesians favour volume; Jerseys are smaller and associated with very high milk fat and hotter conditions. Other supplied breeds offer different composition, adaptation, grazing and temperament combinations. Start with the enterprise goal, comparing benefits and limitations rather than assuming universal suitability.

Functional anatomy connects body parts with their work. The muzzle supports intake; ribs and barrel relate to digestive capacity. Back and loin provide evidence of structural soundness, the body's capacity for support. Udder and teats support milk production and exit; hooves support mobility. Combine relevant tendencies with individual evidence and uncertainty. Inspection, health, welfare and care decisions remain teacher-controlled.



Figure 6. Locate external cow structures; this illustration cannot establish breed, yield, health or welfare.

MODULE 16 / DAIRY FARMING

Explain and check

Written evidence

Hypothetical goal for response 1: a dairy enterprise wants high-protein milk from calm, hardy cattle. Compare Brown Swiss and Holstein-Friesian using the supplied breed tendencies, while recognising that no description guarantees individual performance.

Compare two supplied dairy breeds and recommend one for a stated production goal.

Written evidence

Choose two features and write one cause-and-effect sentence for each, linking its function to production or mobility without diagnosing a condition.

Explain how two external anatomical features can affect a dairy cow’s production or mobility.

Written evidence

Begin your explanation here. Continue the complete source response in your class folio labelled M16-S03. Include every comparison or reasoning step requested.

A dairy enterprise wants a breed with a general tendency towards high milk fat and suitability for hotter conditions. Use breed and anatomy evidence to explain how a decision could be justified without assuming every animal will be identical.

Quick check

Why is the dairy industry important beyond producing milk?

- ○ A. It supports jobs across farming, transport, processing and retail
- ○ B. It operates only in one Australian state
- ○ C. It produces food only for export markets
- ○ D. It removes the need for other agricultural industries

MODULE 17 / DAIRY FARMING

Milk processing, marketing and industry change

From raw milk to a consistent product

Separation divides fat-rich cream from skim milk, allowing fat composition to be adjusted. Pasteurisation uses controlled heating and cooling to reduce harmful bacteria and increase shelf life. It does not separate cream or stop a cream layer forming, because it addresses bacteria rather than fat distribution.

Homogenisation breaks larger fat globules into smaller particles distributed more evenly, so cream does not rise as a separate layer.
It does not remove fat or perform pasteurisation's bacterial-reduction role. The processes complement one another because adjusting fat amount, reducing bacteria and distributing fat are separate objectives. Explain each through starting material, change and result. Equipment settings, hygiene and any practical processing activity require teacher confirmation.

Value adding and dairy marketing

Value adding increases milk's worth through changed form, characteristics or presentation. Physical processing changes texture, form or concentration: churning cream makes butter, while mixing and freezing contribute to ice cream. Microbial processing uses fermentation or biochemical change: bacterial cultures produce yoghurt, while fermentation and enzyme action contribute to cheese.

Higher value does not guarantee profit; costs, quality requirements, demand, capacity and market access matter. Package colours, images, words and slogans support cautious audience inferences. Bright, playful designs may suggest children or families;
nutrition language may suggest health-conscious buyers. Explain the visible evidence using words such as "may", rather than claiming certainty. Lesson packages are examples, not endorsements; current claims need verification.

Deregulation and dairy industry change

Deregulation removes government controls. Australia's 2000 dairy change removed milk-price and quota controls, increasing supply-and-demand influence on farm-gate prices paid to producers. Source effects include price pressure, efficiency demands, larger export-focused farms, processor consolidation—businesses combining operations—and product diversity. Farms differed in scale, costs, location and buyer access, so adaptation varied.

Small-farm losses and financial stress could contribute to rural depopulation and reduced community services. Fewer families may reduce demand for schools and shops. Distinguish policy cause, possible effects and supporting evidence. A balanced judgement considers gains, losses, groups and uncertainty. The 2018 and 2018–2019 datasets are historical, so cannot establish present prices, exports or profitability; current claims need verification.

Different processes have different purposes

Separation	Pasteurisation	Homogenisation
Adjusts fat composition	Reduces harmful bacteria	Distributes fat evenly

Concept organiser based on this module's readings.

MODULE 17 / DAIRY FARMING

Explain and check

Written evidence

Use one concise sentence per process to connect its change with its purpose, keeping the explanation conceptual rather than procedural.

Explain how separation, pasteurisation and homogenisation make different changes to raw milk.

Written evidence

Define each processing type and explain one source example of each, using three or four concise sentences without giving a practical method.

Explain the difference between physical and microbial processing in dairy value adding. Use one source example of each.

Written evidence

Link one farm-context factor with an economic impact and a possible community effect, then acknowledge variation in three or four sentences.

Explain why deregulation affected Australian dairy farms and regions differently. Use one economic factor and one social effect in your response.

Quick check

How should the worksheet’s 2018 and 2018–2019 datasets be used?

- ○ A. As instructions for running a dairy farm
- ○ B. As proof of current 2026 market conditions
- ○ C. As historical evidence for practising data interpretation
- ○ D. As guaranteed forecasts for future prices

MODULE 18 / DAIRY FARMING

Female reproduction and hormones

Female anatomy and fertilisation

Ovaries produce egg cells and hormones; ovulation releases an ovum. The infundibulum collects it into the oviduct, whose upper region is the usual fertilisation site where sperm and ovum join. Oviducts connect with uterine horns and the uterine body. The uterus supports implantation and foetal development, distinct from ovulation and fertilisation.

The cervix protects the uterus; the vagina receives semen and forms part of the birth canal. The vulva is the external opening. Birth passes through these structures. Gestation spans conception to birth; the oestrous cycle repeats reproductive changes. Cattle can cycle repeatedly throughout the year; standing heat means standing to be mounted. Source averages are 283 days for gestation, 21 days for the cycle and about 15 hours for standing heat; individuals vary. Reproductive work remains teacher-controlled.

Reproductive technologies and fertility evidence

Artificial insemination introduces semen at an appropriate reproductive time. Embryo transfer moves a fertilised embryo from donor to recipient; multiple ovulation embryo transfer can involve several embryos. Oestrous synchronisation coordinates cycles using hormones. These purposes depend on timing, health and qualified oversight, without providing reproductive procedures or hormone-use instructions.

In Daisy's source case, condition, heat detection, health and timing may interact. An inference cautiously interprets reported observations; it is not proof of one cause. Earlier reproductive success does not resolve present evidence. Identify relevant facts, explain possible interactions and state uncertainty without blame, diagnosis or treatment advice. Further decisions require qualified health assessment and teacher-controlled management, rather than reliance on a single explanation.

Hormones and the oestrous cycle

The cycle links pro-oestrus, oestrus, metoestrus and dioestrus. FSH, pituitary follicle-stimulating hormone, promotes follicle growth. Growing follicles produce oestrogen, preparing the tract and linking heat behaviour with the LH surge. LH, luteinising hormone, triggers ovulation. The remaining follicle becomes the corpus luteum, an ovarian structure producing progesterone during metoestrus.

Progesterone supports the uterine lining and pregnancy while suppressing heat during dioestrus. Without pregnancy, uterine prostaglandin breaks down the corpus luteum; progesterone falls and follicle growth restarts. Explain the linked changes, not hormone names alone. Synchronisation can coordinate heat detection and breeding management but still depends on health, nutrition, records and professional control. Products, doses, schedules and local reproductive procedures require teacher confirmation.

Classroom evidence case

Source case: Daisy is slightly underweight after calving, shows weak and short heat signs, has a minor uterine infection, was inseminated late in the cycle and produced a normal calf eight months ago.

MODULE 18 / DAIRY FARMING

Explain and check

Written evidence

Use a short linked pathway naming each structure and its function, and clearly identify the usual fertilisation site without practical instructions.

Explain the pathway of an ovum from its release until it reaches the uterus, and identify the usual fertilisation site.

Written evidence

Use three case observations, explain one possible interaction, and distinguish it from certainty before stating the need for qualified assessment.

Using the Daisy case, explain why a fertility decision should consider several pieces of evidence rather than one assumed cause.

Written evidence

Begin your explanation here. Continue the complete source response in your class folio labelled M18-S03. Include every comparison or reasoning step requested.

Explain the hormone sequence from follicle growth to the restart of the oestrous cycle when pregnancy does not occur.

Quick check

Which statement best explains reproductive synchronisation?

- ○ A. It permanently changes the order of the four cycle stages.
- ○ B. It guarantees successful pregnancy regardless of management.
- ○ C. It removes the need for heat detection and records.
- ○ D. It may coordinate heat detection and breeding management, but success still depends on sound management.

MODULE 19 / DAIRY FARMING

Male reproduction and mammary function

Male anatomy and the sperm pathway

Testes produce sperm within the scrotum, which helps maintain suitable conditions. The epididymis is a coiled structure where sperm mature and are stored. The vas deferens transports sperm onwards. Each performs a distinct, connected function, so location alone cannot explain the reproductive pathway.

Accessory glands add fluids to form semen, the mixture of sperm and fluids supporting transport. The penis deposits semen; the sheath protects it when not extended. Follow production, maturation, transport, fluid addition and deposition to explain how the system works together. Sperm cells are not the same as semen. Fertility and monitoring may be researched, but bull handling, collection, testing, diagnosis and treatment are outside student instructions and require teacher confirmation.

Hormones, heat and bull fertility

Pituitary FSH supports spermatogenesis, sperm production; LH, luteinising hormone, stimulates testosterone and supports testicular function.
Testosterone supports sperm formation, reproductive-tract maintenance and libido, or sex drive. It also influences musculoskeletal and secondary sexual development. Visible masculinity does not prove fertility because several functions must work together.

The scrotum keeps testes cooler than body temperature, linking
excessive heat with possible sperm-quality changes. Reduced mating interest after hot conditions records observations, not proof of heat damage or hormonal problems. Libido, discomfort, health and environment may interact. Separate observations, biological relationships and remaining uncertainty; changed breeding results cannot identify a single cause. Further investigation, care and any practical activity need appropriate supervision and teacher confirmation.

Milk formation, let-down and colostrum

The udder has four quarters, each with milk-producing tissue and a teat. Alveoli are sacs where cells use blood-delivered nutrients to form milk. During let-down, oxytocin triggers surrounding myoepithelial cells to contract, moving existing milk through ducts, gland and teat cisterns, then the teat canal. Let-down moves milk; it does not instantly manufacture it.

Ligaments support the udder; the sphincter keeps the teat canal closed between releases. Stress may disrupt let-down. Colostrum is the first secretion around calving, containing immunoglobulins, antibodies providing early immune protection. The source describes declining quality and antibody absorption after birth. These changes matter to early immune protection. Colostrum management, milking, diagnosis and treatment remain teacher-controlled.

Milk formation is not milk let-down

Formation	Let-down
Specialised cells use blood-borne nutrients to form milk	Oxytocin causes surrounding cells to contract and move existing milk

Concept organiser based on this module's readings.

MODULE 19 / DAIRY FARMING

Explain and check

Written evidence

Choose three connected structures and give each function in a short sequence, showing the links with first, next and then.

Describe how three male reproductive structures have different but connected functions.

Written evidence

State the observations, explain two relevant biological links and one uncertainty, using cautious language without diagnosing or recommending tests or treatments.

A bull shows reduced mating interest after a period of hot conditions. Explain how this evidence should be interpreted cautiously.

Written evidence

Explain colostrum's immune value and the general changes after birth in three concise sentences, avoiding numerical claims and farm procedures.

Explain why colostrum is important to a newborn calf without describing a farm procedure.

Quick check

Which statement correctly distinguishes sperm from semen?

- ○ A. Sperm and semen are two names for the scrotum
- ○ B. Sperm are reproductive cells, while semen contains sperm and glandular fluids
- ○ C. Semen is produced only in the testes and contains no sperm
- ○ D. Sperm are fluids produced by the accessory glands

MODULE 20 / DAIRY FARMING

Dairy production, technology and careers

The dairy production cycle

The cycle links calving, lactation, breeding, drying off and sometimes culling. Lactation means milk production, beginning after calving and rising towards a peak. Nutrition also supports maintenance, recovery and reproduction, linking output with condition. Breeding considers reproductive evidence, health history, age, performance and herd goals; successful breeding leads towards another calving.

Drying off ends lactation before calving, allowing udder recovery. Culling removes a cow from the milking herd and may follow persistent fertility, health, production or suitability problems. One observation cannot establish persistence. Compare production records, body condition, health and reproductive information over time to understand connected stages. Source timings are examples, not universal schedules. Local herd choices and care require teacher confirmation.

Selecting dairy technologies using evidence

Dairy technology collects information or provides automation, performing repeated tasks consistently. EID links animals with records; collars and pedometers monitor movement; milk sensors record characteristics; drones show broader paddock conditions. Automated milking, feeding and teat-spray systems support recurring work. Match the technology to the decision, rather than novelty.

Data quality concerns accuracy, completeness and agreement with observations. Missing readings, false alerts or conflicts weaken conclusions. Other limits include cost, maintenance, training, connectivity and integration. Farm context includes herd size, labour, infrastructure, goals and capacity to maintain and interpret systems. Compare functions, benefits, limitations and evidence; no technology is best for every farm. Current performance, compatibility, availability and costs need verification, and local use remains teacher-controlled.

Careers across the dairy supply chain

The dairy supply chain connects careers across production, services, transport and processing. Farmers, herd managers and assistants coordinate on-farm work; veterinarians, nutritionists and reproductive technicians provide specialist support. Equipment technicians maintain technology; data analysts interpret sensors and records. Tanker drivers deliver milk to processors, whose workers contribute to processing, packaging and quality assurance, checking required product qualities.

Advisers and extension officers communicate information. Transferable skills are abilities useful across roles, including teamwork, observation, organisation, communication, problem-solving and technology skills. Compare responsibilities, workplaces, equipment, advantages and challenges with personal interests and demonstrated skills. One role’s output supports another stage, so titles alone cannot explain suitability or connections. Current qualifications, entry requirements and opportunities require authorised confirmation.

Classroom evidence case

Hypothetical record: a movement-monitoring system has no readings for one period and flags reduced activity in another. A visual observation reports normal movement during that second period. Treat the gap and disagreement as evidence-quality concerns, not a diagnosis.

MODULE 20 / DAIRY FARMING

Explain and check

Written evidence

Name two stages, explain their relationship and identify two useful records or observations, keeping the response to three or four sentences.

Explain one relationship between two stages of the dairy production cycle and identify evidence that could inform a decision.

Written evidence

Begin your explanation here. Continue the complete source response in your class folio labelled M20-S02. Include every comparison or reasoning step requested.

Evaluate a set of technology data that contains missing readings or results that conflict with visual observations.

Written evidence

Choose three roles from different areas; briefly state each responsibility and show how their work connects along the supply chain.

Explain how three different dairy careers are connected within the supply chain.

Quick check

Two farms are considering the same automated system. Why might they reach different decisions?

- ○ A. Farm context, infrastructure, labour and goals may differ
- ○ B. Only herd colour should determine the decision
- ○ C. Comparative evidence is unnecessary
- ○ D. Technology has exactly the same value in every setting

DAIRY / TECHNICAL EXPLANATION

Connect anatomy and milk formation

External structures help explain body function. Internal biological processes explain how milk forms and moves. Keep these relationships distinct from animal-care procedures.

A source-grounded mammary pathway

Formation	Movement	Exit
Alveoli: specialised cells form milk using blood-borne nutrients	Myoepithelial cells contract; milk moves through ducts to gland and teat cisterns	Milk leaves through the teat canal

A common misconception

Oxytocin supports milk let-down by causing surrounding cells to contract. It moves milk already formed; it does not instantly create milk.

Draw a conceptual pathway from alveoli to the teat canal. Label the structures and show movement with clear arrows. This is not an anatomical scale drawing.

Explain why formation and let-down are different. Connect one external cow structure to its function and identify an evidence limit.

Explain how colostrum supports early calf immune protection. State why timing matters without prescribing an animal-care procedure.

COURSE SYNTHESIS / EVIDENCE AND NEXT STEPS

Review the whole agricultural course

Use examples from this workbook and your class folio. Evaluate the strength of your evidence, not just whether a page is complete.

Course strand	Your evidence reference	What it demonstrates
Systems and management		
Pastures and cropping		
Prime lamb production		
Dairy farming		

Compare one plant and one animal enterprise using productivity, sustainability, profitability and risk. Explain a justified trade-off.

Choose a decision you changed after checking evidence. Explain what made the revised judgement stronger.

Identify one agricultural career to investigate. Link a skill from this course to the work and state a useful next source or question.